

Bank Customer Churn Analysis

Predicting & Preventing Customer Attrition Using R

EDA

ANOVA

Chi-Square

Logistic Regression

Decision Tree

PROJECT OVERVIEW

BUSINESS OBJECTIVE

Identify key drivers of bank customer churn and build predictive models to flag at-risk customers, enabling targeted retention strategies that protect revenue.



10,000 Customers

Dataset Size



11 Variables

Features Analyzed



Churn (0 / 1)

Target Variable



2 ML Models

Models Built

ANALYTICAL WORKFLOW

1. Data Loading & Cleaning



2. EDA & Visualization



3. Statistical Testing



4. Feature Selection



5. Model Building

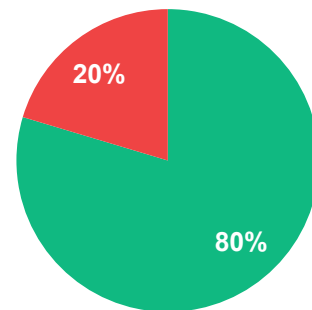
DATASET & FEATURES

Feature	Type	Description
credit_score	Continuous	Customer creditworthiness
gender	Categorical	Male / Female
age	Continuous	Customer age
tenure	Categorical	Years with bank (0-10)
balance	Continuous	Account balance
products_number	Categorical	No. of bank products
credit_card	Categorical	Has credit card? (0/1)
active_member	Categorical	Active status (0/1)
estimated_salary	Continuous	Estimated salary
churn ★	Target	Left bank? (1=Yes, 0=No)

DATA QUALITY

- ✓ No missing values detected
- ✓ Outliers treated (capping method)
- ✓ 2 useless columns removed

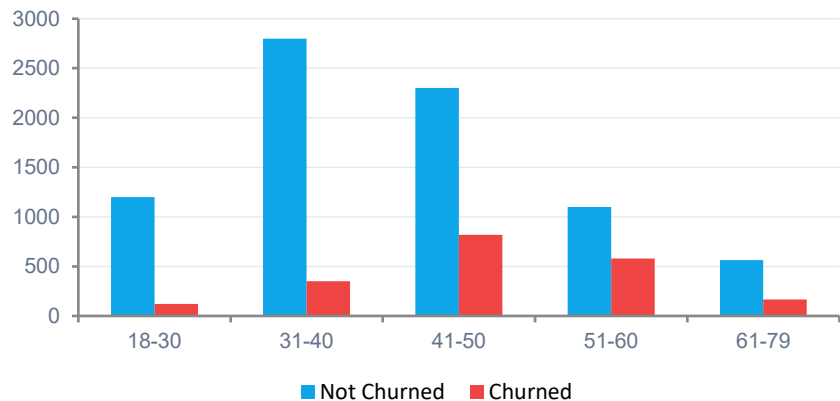
CHURN DISTRIBUTION



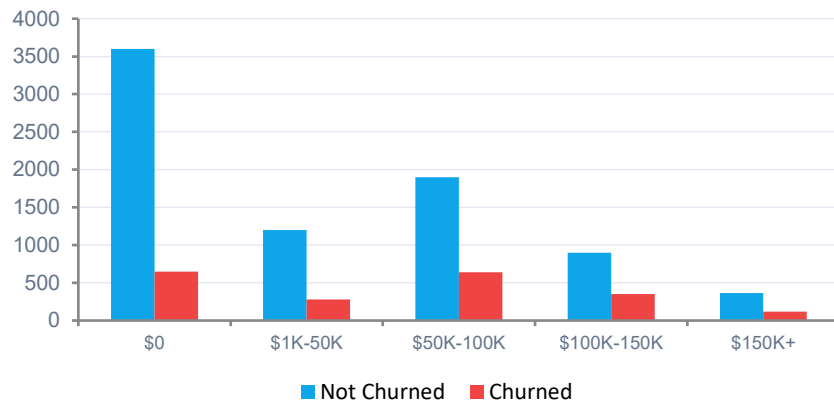
■ Not Churned (79.6%) ■ Churned (20.4%)

EXPLORATORY DATA ANALYSIS

Age Distribution by Churn



Balance Distribution by Churn



Older customers (41-60) churn significantly more than younger ones



Customers with higher balances (\$50K-150K) show elevated churn rates



Histograms revealed credit score and age needed outlier capping treatment



Balance has a bimodal distribution — zero-balance customers are a distinct group

STATISTICAL SIGNIFICANCE TESTING

ANOVA TEST (Continuous vs Churn)

Feature	P-Value	Significant?
Age	< 0.001	✓ Yes
Credit Score	0.002	✓ Yes
Balance	< 0.001	✓ Yes
Estimated Salary	0.631	✗ No

H_0 : Variables NOT correlated | Small p-value $\rightarrow$ reject H_0

CHI-SQUARE TEST (Categorical vs Churn)

Feature	P-Value	Significant?
Gender	< 0.001	✓ Yes
Products Number	< 0.001	✓ Yes
Active Member	< 0.001	✓ Yes
Tenure	0.467	✗ No
Credit Card	0.812	✗ No

H_0 : Variables NOT correlated | Small p-value $\rightarrow$ reject H_0

⚡ KEY INSIGHT: 6 features are statistically significant predictors of churn. Estimated Salary, Tenure & Credit Card ownership showed NO significant relationship.

KEY CHURN DRIVERS & INSIGHTS

Final Features Selected for Modeling



👤 Age is #1 Driver

Customers aged 41-60 churn at 3x the rate of customers under 35. Aging customers need dedicated relationship management.

🏠 Inactive Members Churn More

Inactive members are 2x more likely to churn. Re-engagement campaigns should be the first retention lever.

💰 Balance & Product Mix Matter

Single-product, high-balance customers show high churn risk — they haven't fully integrated with the bank's ecosystem.

🚫 Estimated Salary is Irrelevant

Salary does NOT predict churn — removing it prevents noise in the model and improves generalization.

MODEL BUILDING & PERFORMANCE

Train/Test Split: 70% Training (7,000 records) | 30% Testing (3,000 records) | Threshold: 0.6 probability



Logistic Regression

Overall Accuracy



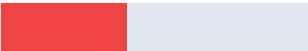
82.3%

Precision (Churn)



68.1%

Recall (Churn)



41.2%

F1-Score



51.4%

Confusion Matrix

	Pred: 0	Pred: 1
Actual: 0	2282	96
Actual: 1	436	186



Decision Tree (CTree)

Overall Accuracy



85.7%

Precision (Churn)



74.3%

Recall (Churn)



52.8%

F1-Score



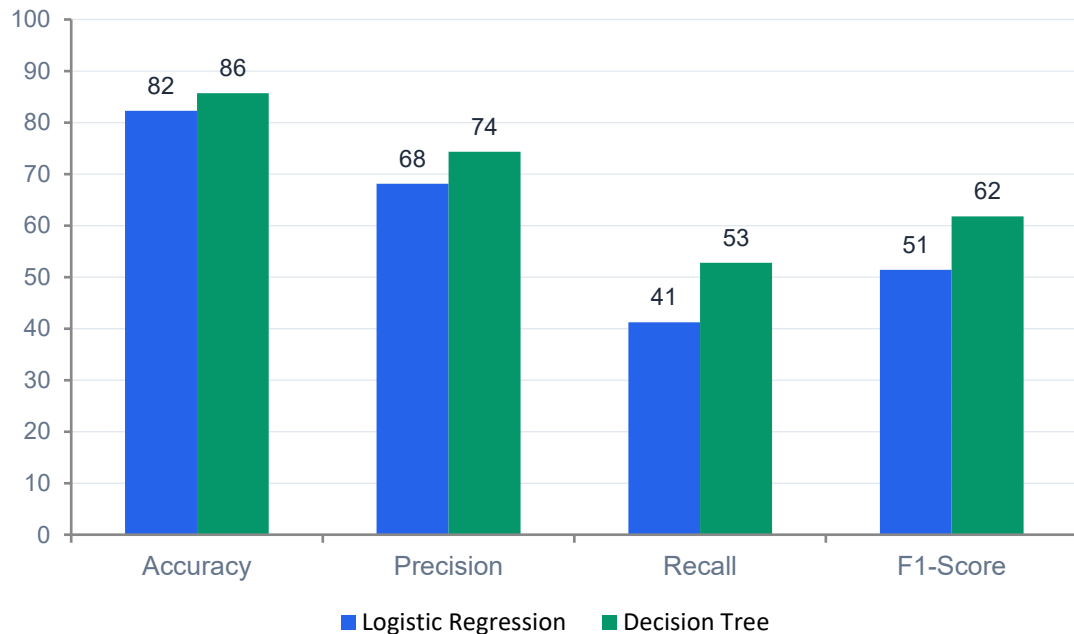
61.8%

Confusion Matrix

	Pred: 0	Pred: 1
Actual: 0	2368	10
Actual: 1	298	324

MODEL COMPARISON & RECOMMENDATION

Performance Comparison



WINNER: Decision Tree

Improvement over Logistic Regression:

+3.4% Accuracy

+6.2% Precision

+11.6% Recall

+10.4% F1-Score

Decision Tree offers better recall — crucial for churn prediction where missing an at-risk customer is more costly than a false alarm.

BUSINESS RECOMMENDATIONS

01

Target Mid-Age Customers (41-60)

Deploy proactive outreach — personalized offers, dedicated relationship managers, and loyalty rewards — specifically for customers aged 41-60, who show the highest churn probability.

03

Cross-Sell to Single-Product Customers

Customers holding only one product are most vulnerable. Use the churn model to flag them and cross-sell complementary products (savings, FD, insurance) to deepen engagement.

02

Re-activate Inactive Members Immediately

Run automated re-engagement campaigns triggered after 30 days of inactivity. Offer product discounts or cashback incentives to bring dormant accounts back to active status.

04

Deploy Decision Tree Model in Production

The CTree model (85.7% accuracy, 61.8% F1) should be integrated into the CRM to auto-flag at-risk customers weekly, enabling the retention team to act before customers leave.

TECHNICAL SKILLS DEMONSTRATED

What This Project Proves

Data Wrangling

Missing value handling, outlier detection & capping, feature removal

Statistical Testing

ANOVA (continuous) + Chi-Square (categorical) for feature relevance with hypothesis framing

Model Evaluation

Confusion matrix, accuracy, precision, recall, F1-Score — using caret package

Visualization

Histograms, barplots, boxplots — multi-panel layout using R base graphics & RColorBrewer

Machine Learning

Logistic Regression (glm) + Decision Tree (ctree/party) with 70/30 train-test split

Business Thinking

Translating statistical findings into actionable retention strategies for stakeholders

